

Jinho Cha, Ph.D.

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RESEARCH PROFILE

Machine Learning • Large-Scale Data Analytics • Statistical Modeling • AI Evaluation • Decision Intelligence • Optimization

Applied AI and data researcher combining machine learning, statistical inference, stochastic modeling, and optimization to build validated decision systems for complex real-world problems. Experienced in leading funded R&D and translating quantitative research into operational impact.

Research Impact: 10 peer-reviewed journal articles published/accepted since 2025 • **\$2.7M+ funded R&D portfolio** across PM/PI/Co-PI roles • 119,390-record ML study with **33.22% realized-profit improvement**.

Selected Publication Venues: *Computers & Industrial Engineering* • *IIE Transactions* • *Journal of Industrial and Management Optimization* • *BMC Public Health*

TECHNICAL SKILLS

- **Core:** Python, R, pandas, NumPy, scikit-learn, LightGBM
- **AI/ML:** PyTorch, TensorFlow, XGBoost, NLP/Text Mining
- **Applied Tools:** Java, Jupyter, SQL, AWS (S3, EC2, SageMaker)
- **Methods:** Predictive Modeling, Statistical Inference, Calibration, Simulation, Robust Optimization, Decision Analytics

SELECTED RESEARCH HIGHLIGHTS

- **Machine Learning & Predictive-Prescriptive Analytics:** Built reservation-level cancellation models on **119,390 hotel bookings** and integrated calibrated risk estimates into rolling booking control, improving mean realized profit by **33.22%**. Published in *Computers & Industrial Engineering*, 2026.
- **Statistical Risk Modeling & Robust Optimization:** Developed Negative Binomial models for clustered, overdispersed risk and distributionally robust sequential control; empirical analyses used S&P 500 jump-count data. Two *JIMO* articles published in 2026.
- **Dynamic Decision Optimization:** Developed stochastic optimization methods for changing quality, latency, reliability, transition costs, drift, and shocks, including stability-aware switching and hysteresis behavior.
- **AI Decision-System Orchestration:** Designed adaptive selection and switching frameworks for heterogeneous decision modules under operational uncertainty, balancing quality, latency, reliability, and switching costs.
- **Simulation & Empirical Decision Analytics:** Built a 10-year stochastic health-policy simulation calibrated with MEPS and NHANES data to evaluate adherence, intervention timing, cost, and ROI. Published in *BMC Public Health*, 2025.

PROFESSIONAL EXPERIENCE

Computer Science Division, Gwinnett Technical College

Faculty, Computer Programming

Conduct AI/data research while teaching programming and computational problem solving; Co-PI on two NITIC-funded AI/education projects (\$20K combined).

Lawrenceville, GA

2025 – Present

Korea Research Institute for Defense Technology Planning and Advancement (KRIT)

Senior Researcher

Led and managed data-intensive R&D spanning analytics, simulation, intelligent systems, logistics, and technology decision support, including a **\$1M national R&D program**.

South Korea

2021 – 2025

Department of Mathematics, Korea Military Academy

Assistant Professor

Conducted quantitative research and taught probability, statistics, linear algebra, and optimization.

Seoul, South Korea

2017 – 2021

Republic of Korea Army Headquarters / Joint Chiefs of Staff

Military Operations Analysis / Operations Research

Applied simulation, operational data analysis, and quantitative decision methods to large-scale mission-critical planning.

South Korea

Prior to 2017

FUNDED R&D LEADERSHIP

- **\$1.0M Project Manager:** Led a national R&D program from problem definition and analytical design through validation and implementation.
- **\$750K Co-PI:** Co-led an intelligent coastal-surveillance program integrating sensing, analytics, AI learning models, and decision support.
- **\$970K+ additional funded R&D:** PI/Co-PI/Co-PM roles in simulation, training analytics, personnel decision systems, and two 2025–2026 NITIC AI-education mini-grants (**\$20K combined**).

EDUCATION & HONORS

Ph.D., Industrial Engineering, Clemson University • **M.S., Industrial Engineering**, University of Florida • **M.S., Industrial Engineering**, Texas A&M University • **B.S., Electronic Engineering**, Korea Military Academy

Presidential Commendation, Republic of Korea (2025) • **Ministerial Commendation**, Ministry of National Defense (2023) • **Army Chief of Staff Commendation** (2019)